

UPPSALA
UNIVERSITET

UPTEC IT 26029

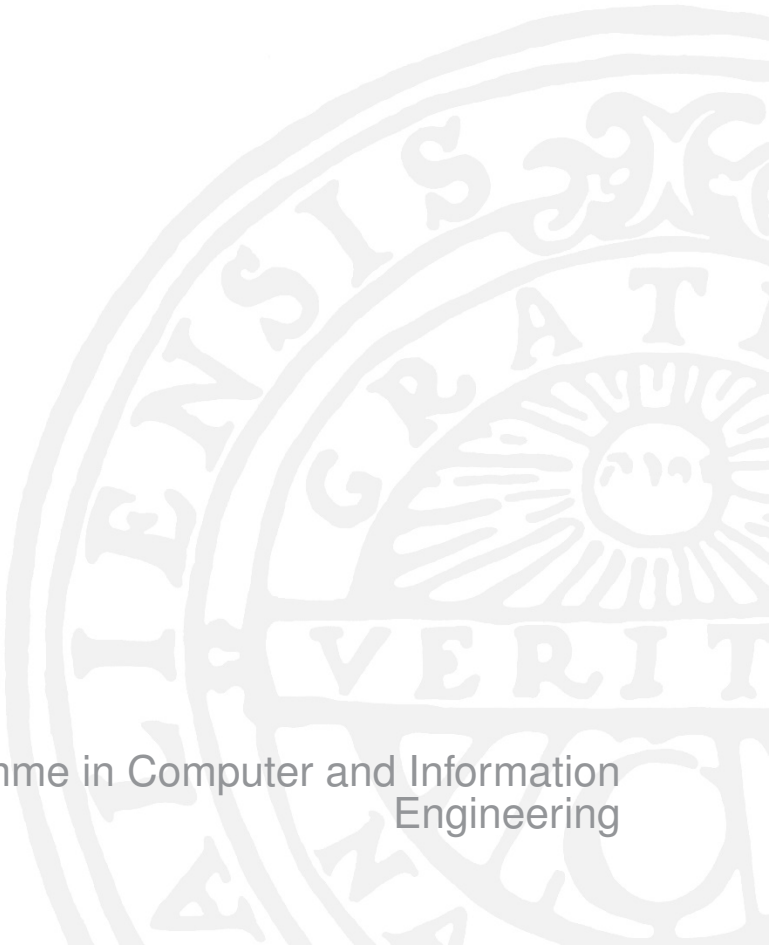
Degree project 30 credits

June, 2026

# Adaptive AI

## Personalization of a Patient-Centered Companion for the Emergency Department Waiting Area

John McLelland



Master's Programme in Computer and Information Engineering



UPPSALA
UNIVERSITET

Adaptive AI

John McLelland


## Abstract

This thesis investigates how a personality-adaptive conversational AI companion can be designed and evaluated for use in Emergency Department (ED) waiting rooms. The aim was to explore whether personality adaptation influences perceived reassurance and usability, and to generate design-relevant knowledge for patient-centred conversational AI in emergency care contexts.

A personality-adaptive AI companion was developed using a Large Language Model (LLM) and personality adaptation based on the Five-Factor Model (FFM). The system was evaluated through a between-groups user study involving ten participants, comparing an adaptive and a non-adaptive version of the companion. Quantitative measures of reassurance and usability were combined with qualitative interviews.

The results showed that the non-adaptive version was perceived as significantly more reassuring than the adaptive version, while no statistically significant difference in usability was observed. Participants associated reassurance primarily with clear explanations, structured guidance, and professional communication. Although the adaptive version produced more conversational and personalised interactions, it was also perceived as more verbose and cognitively demanding.

The findings suggest that personality adaptation alone does not necessarily improve user experience in emergency care settings. Instead, clarity, brevity, and structured communication appear to be key factors influencing reassurance and usability. As an exploratory study conducted in a simulated environment, the findings are intended to generate design insights rather than generalizable conclusions. The study contributes design recommendations for the development of patient-centred conversational AI systems in healthcare.


**Faculty of Science and Technology**

**Uppsala University, Place of publication Uppsala**

Supervisor: Jacobe Klein  Subject reader: Jessica Lindblom

Examiner: Anders Arweström Jansson

# Sammanfattning

Denna studie undersöker hur ett personlighetsanpassad konversationsbaserad AI-system kan utformas och utvärderas för användning i väntrum på akutmottagningar. Syftet var att undersöka om personlighetsanpassning påverkar upplevd trygghet och användbarhet samt att generera designrelevant kunskap för patientcentrerade AI-system inom akutsjukvården.

En personlighetsanpassad AI-kompanjon utvecklades med hjälp av en stor språkmodell (LLM) och personlighetsanpassning baserad på Femfaktormodellen (FFM). Systemet utvärderades genom en experimentell studie med tio deltagare där en adaptiv version av AI-kompanjonen jämfördes med en icke-adaptiv version. Studien kombinerade kvantitativa mätningar av upplevd trygghet och användbarhet med kvalitativa semistrukturerade intervjuer.

Resultaten visade att den icke-adaptiva versionen upplevdes som signifikant mer trygghetsskapande än den adaptiva versionen, medan ingen statistiskt säkerställd skillnad kunde observeras gällande användbarhet. Deltagarna förknippade främst trygghet med tydliga förklaringar, strukturerad vägledning och professionell kommunikation. Även om den adaptiva versionen genererade mer personliga och samtalsliknande interaktioner upplevdes den också som mer omfattande och kognitivt krävande.

Resultaten tyder på att personlighetsanpassning i sig inte nödvändigtvis förbättrar användarupplevelsen i akutsjukvårdskontexter. I stället framstår tydlighet, kortfattadhet och strukturerad kommunikation som centrala faktorer för att skapa både trygghet och god användbarhet. Studien bidrar med designrekommendationer för utvecklingen av patientcentrerade konversationsbaserade AI-system inom hälso- och sjukvården.

**Acknowledgement** I would like to sincerely thank my supervisor, Jacobe Klein, for their guidance and support throughout this project. Their knowledge and encouragement have been greatly appreciated and have helped shape this work from start to finish. I would also like to thank my subject reviewer, Jessica Lindblom, for her insightful comments and constructive feedback, which have helped improve the quality of this thesis.

A special thank you goes to my family for their constant support throughout my studies. Finally, I would like to thank my best friend for encouragement during both the challenging and rewarding moments of this journey. Tack Kevin!

# Contents

**Acronyms** **vii**

**1 Introduction** **1**

1.1 Purpose and Goals . . . . . 2

1.2 Scope and Delimitations . . . . . 2

**2 Background** **4**

2.1 Emergency Departments and Patient Needs . . . . . 4

2.2 Artificial Intelligence . . . . . 5

2.3 Personality traits . . . . . 6

2.4 User Experience . . . . . 7

**3 Method** **9**

3.1 Context and Requirements . . . . . 9

3.2 Design and Implementation . . . . . 10

3.2.1 System overview . . . . . 10

3.2.2 Simulating Conversation . . . . . 11

3.2.3 Personality Computing . . . . . 12

3.2.4 Adapting Personality . . . . . 12

3.3 Empirical Evaluation . . . . . 13

3.3.1 Study design . . . . . 13

3.3.2 Participants . . . . . 14

3.4 Assessment Dimensions . . . . . 15

3.4.1 Performance Evaluation . . . . . 15

3.4.2 Transcript Analysis . . . . . 15

3.4.3 Quantitative Measures . . . . . 16
3.4.4 Qualitative Insights . . . . . 17
3.4.5 Methods of Analysis . . . . . 17

**4 Results** **19**

4.1 Technical Evaluation of the Artefact . . . . . 19
4.2 Empirical Evaluation: Quantitative Findings . . . . . 20
4.3 Empirical Evaluation: Qualitative Findings . . . . . 22
4.3.1 Reassurance . . . . . 23
4.3.2 Satisfaction . . . . . 26
4.3.3 Efficiency . . . . . 27
4.3.4 Effectiveness . . . . . 28

**5 Discussion** **30**

5.1 Key Findings . . . . . 30
5.2 Perceived Reassurance . . . . . 31
5.3 Perceived Usability and User Experience . . . . . 32
5.4 Design Implications . . . . . 33
5.5 Interaction Characteristics and User Perception . . . . . 34
5.6 Methodological Considerations and Limitations . . . . . 34
5.6.1 Personality Computing . . . . . 35
5.6.2 Selection of LLM . . . . . 35
5.6.3 Prompt Design and Personality Adaptivity . . . . . 36
5.6.4 Study design . . . . . 36
5.7 Ethical Considerations . . . . . 37
5.8 Future Work . . . . . 37

**6 Conclusion** **39**

**References** **40**

**A Prompt Engineering Iterations** **44**

A.1 First Iteration . . . . . 44

A.2 Second Iteration . . . . . 44

A.3 Final Iteration . . . . . 46

A.4 Base prompt . . . . . 47

**B Quantitative Questionnaire** **49**

B.1 CUQ . . . . . 49

B.2 PROM-ED . . . . . 52

**C Guiding Interview Questions** **52**

# Acronyms

**AI** Artificial Intelligence. i, ii, iv, 1–5, 8–11, 13, 17, 24, 30–37

**API** Application Programming Interface. 10, 11

**ATI** Attitudes Toward Technology. 13

**BFI-44** 44-item personality assessment. 12, 13

**CA** Conversational Agent. 1, 5–9, 11, 30, 31, 37

**CASA** Computers are Social Actors. 1, 30, 37

**CUQ** Chatbot Usability Scale. 16, 17, 20

**DSR** Design Science Research. 9

**ED** Emergency Department. i, 1–4, 9, 30, 31, 33, 34, 39

**FFM** Five-Factor Model. i, ii, 6, 10, 12

**GenAI** Generative AI. 5

**GWDG** Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. 11

**LLM** Large Language Model. i, ii, v, 2, 3, 5, 9–12, 35, 36

**MBTI** Myers–Briggs Type Indicator. 12

**PACA** Personality-Adaptive Conversational Agent. 1, 7, 37

**UX** User Experience. 1, 7, 8

# 1 Introduction

Emergency Departments (EDs) are characterized by high-pressure environments where patients frequently experience stress and uncertainty. Patients often lack knowledge about the severity of their condition, expected waiting times, and treatment processes, which can lead to increased psychological distress [5, 6]. In such contexts, ensuring that patients feel reassured, informed, and acknowledged is essential for improving their overall experience.

With Artificial Intelligence (AI) being increasingly integrated into daily life and professional workflows, it is crucial to design these systems with a focus on the User Experience (UX), functionality, and robustness. Despite rapid technological advancements, many AI-driven healthcare solutions still fail to place patients at the center of their design [5]. Instead, individuals are often treated as components within a workflow rather than as people navigating vulnerable and uncertain situations.

Conversational Agents (CAs) have shown potential to support users in emotionally demanding situations by acting as social companions and providing interactive support [29, 30]. In healthcare settings, particularly in EDs, such agents could offer reassurance during waiting periods and help alleviate anxiety [13, 30]. Research in human–computer interaction has long shown that people often respond to computers in surprisingly social ways, even when they are fully aware that the system is not human. Studies within the Computers are Social Actors (CASA) paradigm demonstrate that relatively simple cues, such as tone or wording, can lead users to interpret systems as social partners and respond accordingly [27]. This idea is closely related to the Media Equation perspective, which suggests that people tend to treat media technologies as if they were real social entities [28]. From this perspective, it becomes more understandable why a CA could influence feelings such as reassurance or emotional support. However, in a high-stakes context like emergency care, users may instead prioritize clear, structured, and professional communication, which highlights the importance of evaluating how such systems are actually perceived in an emergency waiting-room setting.

Designing effective CAs requires careful consideration of UX, particularly in sensitive contexts such as healthcare. One important aspect is the incorporation of psychological characteristics, including personality traits and emotional expression, which can make interactions feel more natural and engaging [24]. Personality traits influence how users perceive and relate to conversational agents, contributing to more human-like and satisfying interactions [7, 24]. This suggests that personality design may play an important role in developing AI systems intended to support patients in emergency care settings.

Personality-Adaptive Conversational Agents (PACAs) extend this concept by adjusting

their personality based on contextual factors and user characteristics [7, 29]. By tailoring interactions to both the situation and the individual, PACAs may provide more personalized and effective support. This adaptability positions PACAs as a promising approach for applications that aim to address users' emotional and psychological needs.

While conversational AI and adaptive systems have been explored in areas such as mental health and social companionship, their application in emergency healthcare contexts remains limited. This thesis contributes to this area by both developing a context-specific artefact and empirically evaluating how it is perceived.

## 1.1 Purpose and Goals

This thesis follows a design-oriented research approach, where a conversational AI companion for ED waiting rooms is designed, developed, and evaluated. The system incorporates personality-informed interaction with the aim of exploring how such adaptive behavior can support patients during waiting periods.

The aim of this thesis is to generate design-relevant knowledge about how conversational AI companions should be shaped to improve user experience in ED contexts, with a particular focus on perceived reassurance and usability. By comparing adaptive and non-adaptive interaction styles, the study seeks to better understand how personality adaptation influences these aspects.

The following research questions guide the study:

**RQ1** What context-specific requirements and constraints should guide the design of a personality-adaptive conversational AI companion for ED waiting rooms?

**RQ2** How can a personality-adaptive conversational AI companion be designed to meet these requirements and constraints?

**RQ3** How is the resulting system perceived in terms of reassurance and usability, and what interaction characteristics help explain these perceptions?

## 1.2 Scope and Delimitations

This thesis was conducted in collaboration with a German research partner at Freie Universität Berlin as part of a broader research initiative exploring the use of Large Language Models (LLMs) in emergency care contexts. As such, the design of the AI

companion is informed by ongoing work within this research domain, particularly regarding patient-facing conversational systems in ED waiting-room scenarios [5].

The intended use of the AI companion developed in this thesis is intended to provide conversational support and collect relevant patient information in emergency department waiting room contexts. The system is not intended to provide medical diagnoses, treatment recommendations, or replace healthcare professionals.

The scope of the user study focuses on participants' perceived reassurance and usability during short-term interactions with the AI companion in a simulated emergency care setting. Due to the limited number of participants and the simulated nature of the study, the findings cannot be generalized to real clinical environments without further investigation. As such, the findings can only provide limited insight into the design of an AI companion within the specific context of this thesis.

From a technical perspective, the system is constrained by the capabilities and limitations of the underlying LLM and the chosen development platform. The conversational behavior relies on prompt-based control rather than fully trained or fine-tuned models, which limits the precision and consistency of personality adaptation.

# 2 Background

This section provides the foundational context for the thesis by introducing key concepts and technologies related to patient needs, Artificial Intelligence, personality traits, and user experience.

## 2.1 Emergency Departments and Patient Needs

Emergency Departments (EDs) are environments with complex workflows designed to treat life-threatening conditions. Patients attend to EDs with diverse symptoms and varying acuity levels, of which a third is actually classified as low-acuity in retrospect [5]. As part of the triage process, many patients are assigned lower priority due to their lower acuity, which contributes to overcrowding in the ED [6]. Overall, staff shortages and increased workloads lead to frustration among both healthcare staff and patients, ultimately affecting the quality of care and the overall patient experience [5].

From the patient's perspective, multiple perceptions contribute to the overall perceived quality of care. Patients value being taken seriously, even when they present with minor injuries. Furthermore, they need respect, emotional support, and clear communication, especially when they find themselves in vulnerable situations, such as in the emergency waiting room. Hence, empathy and reassurance from clinical staff are crucial in moments like these. In this context, reassurance can refer both to emotional support and to providing clarity, understanding, and a sense of control in an uncertain situation. At the same time, systemic challenges can place patients in a passive role during waiting times, which reinforces feelings of powerlessness and contributes to a sense of disconnection from the care process [5]. In addition to this, patients' needs are dynamic and highly individual, making it difficult for static solutions to the problem. This highlights the potential of implementing adaptive technologies that can complement gaps in human-delivered care. In particular, advancements in Artificial Intelligence (AI) open up new opportunities to support patients during periods of waiting, both functionally and socially [14].

In this context, conversational systems emerge as a particularly relevant form of AI, as they enable direct interaction with patients through natural language. This makes them suitable not only for providing information but also for supporting patients on a social and emotional level during waiting periods.

## 2.2 Artificial Intelligence

AI is a term that was first introduced at a conference in 1956 [33, 15], but it was not until a world chess champion was defeated by an AI program that AI gained global attention [33]. In general, AI is a field of research focused on systems imitating human intelligence, including traits such as learning, judgment, and making decisions [33]. Since the introduction of the term, AI has made significant improvements in various of applications used in different contexts. Among other contexts, AI is used for speech recognition, image processing, natural language processing, and autonomous vehicles [33, 22].

Recent advances have led to an increased use of Generative AI (GenAI) [17, 13], which is a form of AI that is focused on generating human-like content such as text, images, and audio [17]. Large Language Models (LLMs) are neural networks trained on large amounts of text to understand and generate natural language. They are typically based on transformer architectures and can be fine-tuned for specific tasks such as question answering or text generation [17]. LLMs are a type of GenAI that are designed to generate text that is human-like based on a given input, often called a *prompt*. As these systems become more advanced, their role is shifting from purely functional tools toward more socially interactive systems, which has led to increasing interest in AI companions.

AI companions are interactive AI-based systems designed to engage users in ongoing social interaction, often through dialogue and emotionally responsive behavior. Unlike task-oriented systems, AI companions focus on providing social and emotional support, fostering a sense of presence and relational continuity over time. Over the past decades, social companionship between humans and AI has increased [13]. In this context, social companionship refers to "a pleasant and accessible relationship with an interactive source, emerging out of the social and emotional investment of a person, which requires a level of trust, compatibility, and familiarity with the source, resulting in feelings of security and general well-being" [11].

Conversational Agents (CAs) are software-based systems designed to interact with humans through natural language, either through text or speech [25]. CAs takes input, such as written text or spoken language, and processes this information to respond with relevant information for the conversation. CAs has existed for several decades, already in the 1960s there were established chatbots that could simulate a text-based conversation [9]. In the last two decades, there has been an increase in the use of CAs for commercial purposes [9, 25]. However, these agents mostly only functioned using constrained user input, not having the capability to understand natural language input [9].

The current rapid growth in abilities introduced by LLMs has enabled CAs to process

natural language and been able to generate better and more relevant responses. As a consequence of this, CAs have increasingly been used in domains such as assistance agents, for educational purposes, and in health care [25]. As CAs increasingly take on roles that involve social interaction, how they communicate becomes just as important as what they communicate. This has led to growing interest in incorporating human-like characteristics, such as personality, into these systems. This could both develop more complete cognitive models of CAs and enable more natural, human-like interactions with users [7].

## 2.3 Personality traits

In the context of conversational systems, personality traits provide a way to shape how interactions are experienced by users. Every individual is different. Some people are more talkative, while others are quieter. Some worry a great deal, while others rarely feel anxious. Using terms as "talkative," "quiet," "active," or "anxious" describes aspects of an individual's personality [12]. To better understand differences in individual personalities, defining personality traits helps capture the basic dimensions along which personalities may vary [12]. Personality traits reflect the characteristics of the thoughts, feelings, and behaviors of individuals. Over time, multiple frameworks have been established to describe personality, including the Big Five Personality Traits model, also referred to as the Five-Factor Model (FFM) [23].

The FFM conceptualises personality as consisting of five broad dimensions: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness to Experience [26]. The model emerged from research based on both natural language descriptions of personality and questionnaire-based psychological assessments, and it has since developed into a scientific standard. A substantial body of research demonstrates that these five factors are comprehensive, relatively stable throughout adulthood, and consistently observable across different cultures and languages. For this reason, the FFM is the most widely used model in both psychology and personality computing [16, 1, 26].

Researchers use the FFM as a common framework to organize personality cues across verbal, paraverbal, and body language. By tweaking things like how much a CA talks, speech speed, or gestures, designers can influence how users perceive its personality. Using the FFM this way could help create a social interaction channel that makes conversations feel smoother and more believable [7]. Applying the framework for personalities is a powerful tool to give CAs more character and improve the overall user experience with CAs. This makes personality a relevant design dimension in conversational systems, as it can influence how users interpret and respond to system behavior.

## 2.4 User Experience

Overall, User Experience (UX) is the users' expectations prior to interaction, their experiences during use, and their reflections after use of an interactive system. In this sense, UX reflects both utility qualities, such as functionality and usability, and subjective qualities, such as enjoyment, stimulation, and satisfaction [8]. UX has emerged as a central concept in the design and evaluation of interactive systems and extends beyond focusing on usability by also addressing how users perceive, feel, and respond to it [8].

Historically, the concept of UX has evolved from research in human–computer interaction and usability engineering, where the primary focus was on optimizing performance and minimizing error [8]. Over time, researchers and practitioners realized that simply making a system functional was not enough to create a good user experience. This led to a broader view of interaction design, having researchers questioning what a good user experience actually is. More attention was given to how users feel, what they expect, and the context in which the interaction takes place. As a result, UX has grown into a field that combines knowledge from areas such as psychology, design, information systems, and marketing [8].

In healthcare contexts, UX refers to the patient's overall experience of the care they receive. Creating a positive user experience is particularly challenging due to the complexity and sensitivity of clinical environments. UX in healthcare goes beyond general usability concerns and directly influences patient safety, trust, and emotional well-being. Poor design can create problems in practice, whereas good design can support safer and more efficient care delivery [25, 10]. The design of digital systems in healthcare is therefore important to take into consideration as it has a direct impact on care.

The design of CAs in health care requires multiple perspectives to be taken into consideration, especially when focusing on UX. By incorporating multiple psychological layers into the design of a CA, the system can be utilized in a manner that allows for a more human-like, natural conversation. These layers could include features such as personality traits and the display of emotion [24]. Personality traits in CAs influence the user's perception of the CA and can lead to a more pleasant conversation, making it a valuable layer to consider when developing CAs for health care. Personality-Adaptive Conversational Agents (PACAs) is a version of CAs that is built to adapt the personality of the CA to the context in which it is used [29]. By adjusting personality traits based on situation and the personality of the user, PACAs could enable a more personalized conversation and possibly contribute to a positive UX. However, the effects of such personalization are not necessarily straightforward. Depending on the context, user expectations, and the nature of the task, more adaptive or expressive interaction may not always lead to a better UX. In environments such as emergency healthcare, users may

instead prioritize clarity, efficiency, and professionalism.

Taken together, prior research shows that people tend to respond to interactive systems in social ways, that conversational agents enable such interactions in practice, and that design choices such as personality can shape how these interactions are perceived. This suggests that AI companions in the form of CAs may influence UX not only through their functionality, but also through how they communicate. At the same time, the effects of such design choices are likely to depend on context, highlighting the need to better understand how these systems are perceived in specific domains.

# 3 Method

The methodological approach adopted in this study combines system requirements, system design and development, and empirical evaluation using a mixed-methods assessment. The work follows a design-oriented research logic, where knowledge is generated through the design, development, and evaluation of a conversational AI artefact in an ED waiting-room context. This approach is aligned with Design Science Research (DSR), in which an artefact is constructed to address a real-world problem and evaluated to generate design-relevant insights [19]. This supports RQ1 and RQ2 by linking requirements to artefact design, while RQ3 is addressed through the empirical evaluation of user interaction and perception.

In line with this, the study involves identifying context-specific requirements, implementing a personality-adaptive CA, and assessing how it is perceived in use. The relevance cycle is grounded in previously identified needs and prior research concerning the use of CAs in the context of EDs. The design cycle focuses on the development of the artefact, including architectural and prompt-level design decisions. Finally, the rigor cycle emphasizes systematic evaluation using established methods and existing theoretical foundations.

A personality-adaptive CA was implemented and subsequently evaluated through a controlled between-group user study. Quantitative and qualitative measures were used to examine perceived usability and reassurance. Before the user study, a basic system-level performance check was conducted to assess response characteristics such as latency and response length, and to identify potential issues that could influence user perception.

The following subsections describe the system design and implementation, the experimental setup and participant procedure, and the evaluation framework used to address the research questions.

## 3.1 Context and Requirements

The relevance of the artefact is grounded in the ED waiting-room context as well as in ongoing research exploring LLM-based patient-facing applications in emergency care. In this setting, patients often experience uncertainty, lack of information, and limited interaction during waiting periods. These conditions highlight both the practical potential of conversational systems in emergency environments and the need for careful design in privacy-sensitive healthcare contexts [5]. Related research on patient-facing companion concepts for ED waiting areas further motivates the focus on how such systems are perceived in this context [5, 14, 25]. Based on these contextual factors, the system was

designed to provide both informational and emotional support, addressing functional as well as experiential requirements.

## 3.2 Design and Implementation

This section describes the technical design and implementation of the system developed. The implementation focuses on three core aspects: the overall system architecture, the integration of personality computing, and the mechanisms used to simulate natural conversation through an LLM.

### 3.2.1 System overview

The system was implemented using a client–server architecture that combined a web-based frontend, a Python backend, and an external LLM-Application Programming Interface (API) to support natural language interaction. As illustrated in Figure 1, user interaction began in the web-based frontend, developed using Vite in combination with a JavaScript UI framework [4]. User input was captured through a chat interface and sent to the backend via an HTTP POST request to a Flask application. Within the backend, user input is incorporated when defining the prompt along with the FFM values of the user. The constructed prompt is sent to an external API, which generates the AI companion's response. Finally, the response is returned through the backend to the frontend, where it is displayed to the user.

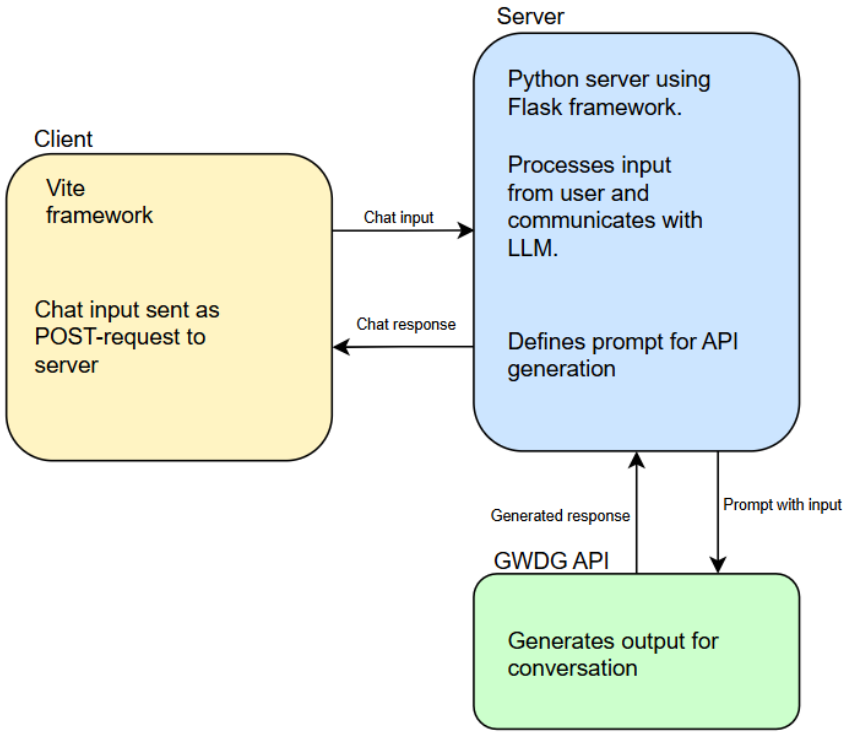



**Figure 1** Overview of the implemented artefact. The system enables a conversational interaction where user input from the client is processed by a server and forwarded as prompts to the GWDG API. The generated responses are returned to the client, supporting real-time interaction.

### 3.2.2 Simulating Conversation

An LLM was central to the implementation of the CA, enabling the simulation of natural conversation. For this purpose, access to the Scalable Artificial Intelligence Accelerator at Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen (GWDG) [2] was granted via an API key. This service provides access to a wide range of hosted large language models, enabling the selection of an appropriate model for the system. As part of the GWDG AI services, multiple models with varying capabilities and parameter sizes were available [3].

When considering different models, it was important to select one capable of understanding relatively complex prompts and simulating natural conversation. This was necessary because prompting served not only as a method for generating coherent responses but also for enabling personality-adaptive output. The model `llama-3.3-70b-instruct` was selected for the system. This choice is motivated

by the perceived strong balance between linguistic quality and computational efficiency in comparison to other options. The rationale for this selection is discussed further in Section 5.6.2.

The decision to use an externally hosted LLM was motivated by both the diversity of available models and the advantage of accessing models with a significantly larger number of parameters. This enabled greater adaptability to prompt engineering and improved the quality and robustness of generated responses.

### 3.2.3 Personality Computing

Personality is a complex construct and cannot be measured directly as a single variable. However, established frameworks enable personality computing across defined dimensions. In this study, the FFM and the MBTI were considered due to their widespread use in personality computing research [16]. After comparing these alternatives, the FFM was selected as the personality framework used in the system.

As described in Section 2.3, the FFM defines personality as five continuous dimensions, yielding the possibility for numerical representation of the personalities of users. This characteristic makes the framework well suited for personality inference in this thesis. In addition, the FFM is strongly grounded in empirical research and has demonstrated high reliability and validity across studies, resulting in extensive documentation, datasets, and existing computational models [26]. These properties provide a robust foundation for personality computing within the scope of this thesis.

### 3.2.4 Adapting Personality

Prompt engineering was used for adapting the personality of the LLM to the personality of the user. Rather than shifting the underlying model weights, user personality were encoded into the system prompt and used to guide response interaction behavior.

Participants' personality traits were collected prior to the experiment using the 44-item personality assessment (BFI-44) [23]. User personality was therefore represented through static FFM trait values, which were incorporated into the system prompt as contextual information. In the prompt, instructions were provided to constrain the model's purpose and boundaries while simulating conversation, while personality-related information was used as a framework for stylistic adaptation. The system prompt was iteratively refined to balance realistic behavior with clear constraints. The iterations focused on preventing exaggerated behavior of the personality adaptation and maintaining a consistent interaction style appropriate for the context of the study. The full set of prompt

iterations is provided in Appendix A.

Using prompt engineering for personality adaptation offered rapid iteration without introducing additional training complexity. Fine-tuning was also considered as an additional feature for the personality adaptation, but was not pursued due to the computational resources required.

## 3.3 Empirical Evaluation

This section covers the methodology of the experimental design, as well as information and demographics of the participants.

### 3.3.1 Study design

To investigate the different perceptions of personality-adaptiveness, the study employed a between-groups experimental design with two conditions: one featuring a personality-adaptive AI companion and one without personality adaptation. The independent variable was the presence of personality adaptation in the AI companion. Participants were randomly assigned to interact with either a system that incorporated personality-adaptive behavior or a baseline (non-adaptive) version without this feature. The dependent variables were perceived usability and perceived reassurance during the interaction. A visualisation of the study procedure can be seen in Figure 2.

Before the experiment, participants completed a demographic questionnaire, a brief measure of Attitudes Toward Technology (ATI) [18], and the BFI-44 [23]. For participants in the group with the adaptive version, this data was used as static input to guide the personality of the AI companion. The main part of the experiment consisted of the interaction with the AI companion. During the interaction, participants were asked to take on the role of patients in an emergency waiting room. They were provided with a digital description of symptoms, scenario, and conversational guidelines, which remained accessible throughout their interaction with the AI companion. The study was not conducted with actual patients in a real emergency setting due to ethical considerations.

The study was conducted digitally, and all participants took part remotely. After completing the interaction, participants answered a questionnaire assessing perceived usability and reassurance, both of which are available in Appendix B. The resulting scores provided comparable measures of usability and reassurance across the two groups. Finally, a semi-structured interview was conducted to gain a broader understanding of

their experiences with the artefact, the guiding questions can be found in Appendix C.

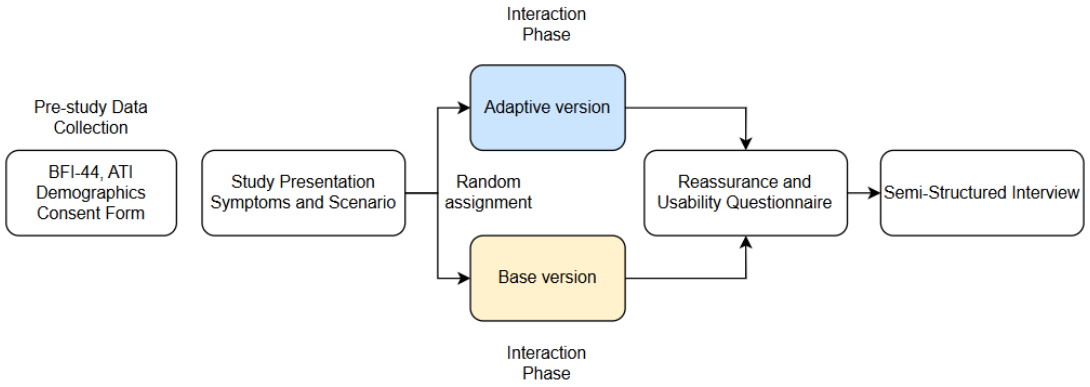



**Figure 2** Study design and procedure

### 3.3.2 Participants

A total of 10 participants were recruited for the study, consisting of 4 male and 6 female participants. As seen in Table 1, participants were distributed across three age groups (18–24, 25–34, and 45–54), with the majority belonging to the 18–24 category. The participants represented a range of educational backgrounds, including upper secondary education, post-secondary education (incomplete), vocational education, and master's degrees.

| Alias | Age | Gender | Education |
|---|---|---|---|
| P1 | 18–24 | Male | Upper secondary education |
| P2 | 18–24 | Female | Post-secondary education (incomplete) |
| P3 | 25–34 | Female | Master's degree |
| P4 | 18–24 | Male | Master's degree |
| P5 | 45–54 | Male | Upper secondary education |
| P6 | 18–24 | Male | Upper secondary education |
| P7 | 18–24 | Female | Upper secondary education |
| P8 | 18–24 | Female | Upper secondary education |
| P9 | 25–34 | Female | Master's degree |
| P10 | 45–54 | Female | Vocational/Adult education |

**Table 1** Participant demographics

## 3.4 Assessment Dimensions

This section outlines the methodology used to evaluate the artefact and address the research questions. The assessment covers performance evaluation, quantitative measurements, and qualitative insights, the latter two focusing on perceived usability and reassurance.

### 3.4.1 Performance Evaluation

The performance evaluation served as a technical assessment of the artefact, complementing the user-centered evaluation by examining system-level trade-offs between responsiveness and response richness.

Both the adaptive and non-adaptive conditions used the same underlying language model `llama-3.3-70b-instruct`. To control for input variability, a fixed set of identical user prompts (N = 8) was used across both conditions. These prompts were manually selected to represent common interaction types in the emergency department context, including symptom descriptions (e.g., “My stomach hurts”) and emotional expressions (e.g., ”I am a bit worried right now”).

The only experimental manipulation was the system-level prompt: the adaptive condition included personality-based and behavioural instructions, while the non-adaptive condition used a simplified prompt.

For each request, response latency was measured as the elapsed time (in seconds) between sending the API request and receiving the generated output. Response length was operationalised as the number of characters in the assistant’s reply. For each condition, average latency, standard deviation, minimum, and maximum latency were computed, along with average response length.

Each prompt was evaluated once per condition under identical technical conditions. As such, the results provide a controlled but limited estimate of system performance and do not capture variability across repeated runs.

### 3.4.2 Transcript Analysis

To complement the technical evaluation, a retrospective transcript analysis was conducted on the conversations generated during the user study. This analysis examined structural and interactional properties of the dialogues to better understand how system behaviour manifested in practice.

The following measures were operationalised:

- **Assistant questions:** total number of question marks (”?”) in assistant messages, used as a proxy for proactive engagement.
- **User repetitions:** number of repeated user messages within a conversation, calculated as the difference between total user turns and the number of unique user messages.
- **Response length:** average number of characters and words per assistant turn.
- **Dialogue length:** total number of turns per conversation.

These measures are descriptive and do not directly capture conversational quality or valence. Instead, they provide an indication of interaction patterns.

### 3.4.3 Quantitative Measures

The quantitative metrics for the experiment focused on measuring the perceived usability and reassurance, as these dimensions are central to assessing the acceptability of interactive systems. Usability is commonly defined as the extent to which a system can be used to achieve specified goals with effectiveness, efficiency, and satisfaction in a particular context of use [21]. Together, these three pillars provide a structured framework for evaluating whether a system supports users in completing tasks accurately (effectiveness), with minimal effort and time (efficiency), and in a manner that is perceived as positive and acceptable (satisfaction).

In addition to functional usability, emotional and psychological responses to technology are particularly important in healthcare contexts [5]. Reassurance refers to users' perceptions of safety, confidence, and emotional comfort when interacting with a system. The perceived reassurance during interaction with a system was deemed relevant for the context of the thesis. Consequently, reassurance was treated as an additional metric alongside usability.

To quantitatively assess these aspects, several established measurement instruments were reviewed. The final questionnaire combined usability-focused items from the Chatbot Usability Scale (CUQ) [20] together with reassurance-oriented items adapted from the PROM-ED framework [31]. This combination enabled a balanced evaluation of both task-oriented system performance and users' subjective feelings of confidence and safety. The complete set of questionnaire items used in the study is provided in Appendix B.

To address RQ3, the quantitative measures of perceived usability and reassurance were compared between the two experimental groups. The analysis focused on identifying differences in mean scores between participants interacting with the personality-adaptive AI companion and those interacting with the non-adaptive system. This approach enabled evaluation of whether the inclusion of personality adaptation in the AI companion had a measurable impact on users' perceived usability and reassurance.

### 3.4.4 Qualitative Insights

In addition to the quantitative measures, a qualitative component in the shape of a semi-structured interview was included to provide a deeper understanding of participants' experiences. The purpose of the semi-structured interview was to capture nuanced insights into how participants perceived and interpreted their interaction with the AI companion. While the questionnaire data enabled systematic comparison between conditions, the interviews allowed participants to elaborate on their responses and highlight aspects of the interaction that may not have been fully captured by standardized measures. The guiding questions were open-ended, allowing participants to reflect on their overall experience of the interaction, as well as their perceived reassurance and usability. These questions are provided in Appendix C.

This qualitative approach served to complement and contextualize the quantitative findings. In particular, it provided insight into why certain patterns may have emerged in the measured usability and reassurance scores. By combining quantitative and qualitative data, the study aimed to achieve a more comprehensive understanding of the interaction and to strengthen the interpretation and validity of the results.

### 3.4.5 Methods of Analysis

The quantitative questionnaire data consisted of responses collected using Likert scales, which were converted into numerical values for analysis. Reassurance-related items were mapped to a numerical scale ranging from 1 to 4. Negatively worded items were reverse-scored prior to analysis to ensure that higher values consistently represented higher levels of reassurance. An average reassurance score was then calculated for each participant.

The usability score was calculated in accordance with the CUQ framework and its recommended scoring procedure [20]. The resulting CUQ scores ranged from 0 to 100, providing a standardized measure of perceived usability. The average scores for each participant were graphically represented using boxplots to visualize and compare dif-

ferences between the study groups.

To statistically assess differences between the adaptive and non-adaptive chatbot conditions, the Mann-Whitney test was used. This non-parametric test was chosen due to the ordinal nature of Likert-scale data and the relatively small sample size, which do not necessarily meet the assumptions of normality required for parametric tests such as the independent samples t-test [32]. The Mann-Whitney test was therefore appropriate for comparing differences in central tendency between the two groups.

The qualitative data were analysed using a deductive thematic approach. The interview material was coded according to the predefined categories of reassurance, effectiveness, efficiency, and satisfaction. The coded extracts were then analysed while taking the between-group structure of the study into consideration.

# 4 Results

To study the artefacts' impact on the users' perceived usability and reassurance, a study with semi-structured interviews was conducted. A total of 10 participants took part in the study. The results consisted of quantitative measurements in the form of average scores of perceived usability and reassurance, alongside qualitative insights and recurring themes derived from the interview data. The demographics, ATI scores, and version of the artefact that was interacted with can be seen in Table 2.

| Alias | Age | Gender | ATI | Adaptive Version |
|---|---|---|---|---|
| P1 | 18–24 | Male | 3.40 | Yes |
| P2 | 18–24 | Female | 3.33 | Yes |
| P3 | 25–34 | Female | 4.17 | No |
| P4 | 18–24 | Male | 4.00 | Yes |
| P5 | 45–54 | Male | 3.38 | No |
| P6 | 18–24 | Male | 4.00 | No |
| P7 | 18–24 | Female | 3.83 | No |
| P8 | 18–24 | Female | 2.50 | No |
| P9 | 25–34 | Female | 1.75 | Yes |
| P10 | 45–54 | Female | 3.62 | Yes |

**Table 2** Participant demographics, ATI scores and which system they evaluated

## 4.1 Technical Evaluation of the Artefact

Table 3 presents the results of the controlled performance evaluation.

| Metric | Adaptive | Base | Difference |
|---|---|---|---|
| Avg. latency (s) | 7.56 | 7.06 | +0.50 |
| Latency std (s) | 1.44 | 2.35 | -0.91 |
| Min latency (s) | 5.83 | 3.98 | +1.85 |
| Max latency (s) | 9.94 | 11.34 | -1.40 |
| Avg. chatbot response length | 159.88 | 141.63 | +18.25 |

**Table 3** Controlled performance comparison between adaptive and non-adaptive prompting

The performance results show that the adaptive prompt produced slightly higher average response latency compared to the non-adaptive. However, the adaptive condition

exhibited lower variability in response times, as indicated by a smaller standard deviation. In terms of output characteristics, the adaptive version generated longer responses on average, reflecting increased verbosity.

To contextualize these findings in real user interactions, Table 4 presents the results of the retrospective transcript analysis.

| Metric | Adaptive | Base | Difference |
|---|---|---|---|
| Turns | 24.4 | 21.2 | +3.2 |
| User turns | 12.2 | 10.6 | +1.6 |
| Assistant turns | 12.2 | 10.6 | +1.6 |
| Avg. chatbot response length (chars) | 419.6 | 357.6 | +62.0 |
| Avg. chatbot response length (words) | 80.45 | 69.42 | +11.03 |
| Assistant questions | 21.0 | 14.2 | +6.8 |
| User repetitions | 0.4 | 0.6 | -0.2 |

**Table 4** Retrospective transcript analysis comparing adaptive and non-adaptive interaction

The transcript analysis shows that the adaptive version led to longer and more interactive conversations, reflected in a higher number of turns and a greater number of assistant questions. Responses were also longer on average, indicating a more verbose conversational style. Additionally, user repetitions were slightly lower in the adaptive condition.

## 4.2 Empirical Evaluation: Quantitative Findings

Perceived usability was evaluated using the CUQ [20] score (0–100 scale). The non-adaptive condition showed higher usability scores overall ($M = 87.50$, $Md = 85.94$) compared to the adaptive condition ($M = 80.63$, $Md = 78.13$).

As shown in Figure 3, the adaptive condition exhibited greater variability, with a wider interquartile range ($IQR = 15.63$; $Q_1 = 73.44$, $Q_3 = 89.06$) compared to the non-adaptive condition ($IQR = 13.28$; $Q_1 = 81.25$, $Q_3 = 94.53$).

The distribution of scores in the non-adaptive condition appears more concentrated towards the higher end of the scale, whereas the adaptive condition shows a broader spread, indicating more variability in perceived usability across participants.

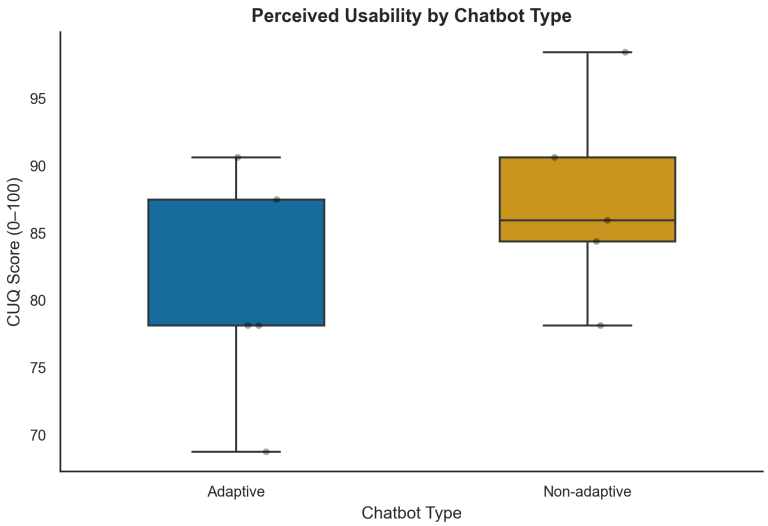



**Figure 3** Distribution of CUQ scores across chatbot conditions. Boxes represent interquartile ranges, medians are indicated by central lines, and individual observations are overlaid.

Perceived reassurance was measured as the mean score across four items [31]. The non-adaptive condition showed higher reassurance scores overall ($M = 3.70$, $Mdn = 3.75$) compared to the adaptive condition ($M = 2.70$, $Mdn = 3.00$).

As shown in Figure 4, the adaptive condition exhibited greater variability, with a wider interquartile range ($IQR = 1.25$; $Q_1 = 2.00$, $Q_3 = 3.25$) compared to the non-adaptive condition ($IQR = 0.62$; $Q_1 = 3.38$, $Q_3 = 4.00$).

The distribution of scores in the non-adaptive condition appears more concentrated towards the higher end of the scale, whereas the adaptive condition shows a broader spread, indicating more variability in perceived reassurance across participants.

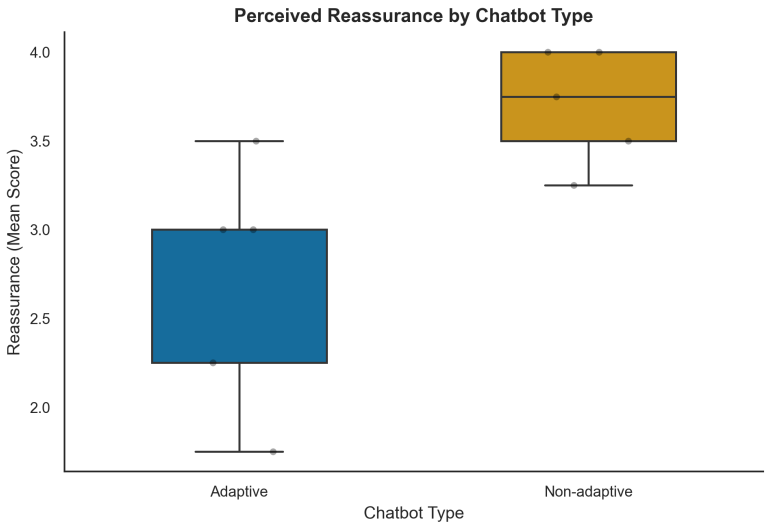



**Figure 4** Distribution of reassurance scores across chatbot conditions. Higher scores were observed in the non-adaptive condition.

A Mann-Whitney test was conducted to assess differences between the adaptive and non-adaptive chatbot conditions. For perceived usability, the difference between conditions was not statistically significant ($U = 7.5$, $p = .34$). In contrast, the difference in perceived reassurance was statistically significant ($U = 1.5$, $p = .027$). Participants interacting with the non-adaptive chatbot reported significantly higher levels of perceived reassurance compared to those interacting with the adaptive chatbot; the results are presented in Table 5.

| Measure | U | p-value | Significance |
|---|---|---|---|
| CUQ (Usability) | 7.5 | .34 | Not significant |
| Reassurance | 1.5 | .027 | Significant |

**Table 5** Results of Mann–Whitney U tests comparing adaptive and non-adaptive chatbot conditions

## 4.3 Empirical Evaluation: Qualitative Findings

The deductive analysis presented in this section is based on the semi-structured interviews conducted after the interaction with the AI companion. All interviews were conducted in Swedish, and all quoted material has been translated into English by the author.

The deductive analysis of the interview data was conducted using reassurance as one code, alongside the three pillars of usability: satisfaction, efficiency, and effectiveness. The frequency of identified codes in the interview transcripts was relatively similar across the different versions that participants interacted with, this can be seen in Table 6. However, it is important to note that this includes both positive and negative statements within each coded category. Code frequencies are reported for transparency; hence, they should not be interpreted as indicators of quality or desirability. Instead, the qualitative findings are based on thematic interpretation.

| Version | Reassurance | Satisfaction | Efficiency | Effectiveness |
|---|---|---|---|---|
| Adaptive | 22 | 39 | 19 | 21 |
| Non-adaptive | 14 | 45 | 13 | 26 |

**Table 6** Frequency of identified codes in the qualitative deductive analysis

The analysis is structured according to the selected codes. Within each section, the key findings are first presented for the adaptive interaction, followed by the findings for the non-adaptive interaction.

### 4.3.1 Reassurance

**Participants in the adaptive interaction** generally described experiencing a sense of reduced worry, emotional relief, and reassurance, often in terms of feeling supported and confirmed in their concerns. For example, P1 described:

> *"...So, yes, I felt a bit calmer. I had a bit less worry. The worry was still there, but it eased somewhat. And you got answers to your questions..." (P1)*

This indicates that while concern was not fully eliminated, it was alleviated. Similarly, reassurance was linked to the feeling of having a conversational partner, as P1 further noted:

> *"Alma made me feel calmer, because you had someone to talk to, you could vent. She listened and gave some perspective and follow-up questions..." (P1)*

while P2, P4 and P10 highlighted that:

*"It felt nice to get that kind of confirmation." (P2)*

*"Very pleasant, informative, and also calming." (P4)*

*"It felt, yes, safe and nice actually." (P10)*

further reinforcing the sense of emotional security reported across participants.

In contrast to this, some participants did not experience the adaptive interaction as reassuring, instead expressing a lack of emotional connection. These participants described that the interaction did not meet their expectations of reassurance, particularly in terms of emotional support or comfort. For instance, P9 expressed a clear skepticism towards the system's ability to provide emotional reassurance and also AI systems in general:

*"...that it should give me something more, like more reassurance... I feel like I don't get anything out of it. I'm not someone who has an AI boyfriend... I don't feel like I get anything out of chatting with AI bots on an emotional level." (P9)*

This highlights a perceived limitation in the system's ability to provide meaningful emotional support. Instead, reassurance was described as being more closely tied to receiving concrete and explanatory information. P9 further elaborated:

*"I think I would have felt calmer if it gave more explanation of what I'm feeling physically... then I know the reason... I don't think it gave me that much... I'm more focused on getting answers about my symptoms... you want to know what it could be physically..." (P9)*

Additionally, some participants indicated that they did not seek or value emotional reassurance from a chatbot at all. P4 noted:

*"...I don't feel that I need that kind of comforting need or sympathy... at least not from a chatbot." (P4)*

Together, these accounts contrast with the previously described experiences by illustrating that reassurance is not uniformly perceived across participants. Instead, for some, the adaptive interaction did not fulfill emotional needs and was primarily evaluated based on its ability to provide clear and concrete information.

**Participants in the non-adaptive interaction** generally described experiencing reassurance in terms of feeling calm, supported, and taken seriously. Many participants emphasized a sense of being listened to and guided through the interaction in a structured and careful manner. For example, P8 and P3 described the experience as highly reassuring:

> *"It felt really good... if it had been real, it would have felt very reassuring. You were able to elaborate, and Alma asked all the questions and went through everything calmly and carefully so that you felt safe and listened to...it wasn't stressful... it was more like 'okay, I hear you... that sounds uncomfortable...' which makes you feel calmer... you get time to reflect and respond, and then receive a professional answer..." (P8)*

> *"I felt seen. You feel safe... safe in that it is being picked up by someone who wants you well... it was a good feeling." (P3)*

Reassurance was also connected to the possibility of early support and reduced stress in an otherwise stressful context. P3 further noted:

> *"...you become more stressed as a patient if you are waiting... Alma could be a safe first encounter... I get to express how I feel... " (P3)*

In addition, participants highlighted how explanations and normalization of symptoms contributed to feeling calmer. P7 described that:

> *"Yes, it definitely made me feel calmer... it explained that it's normal to feel pain... and that makes you feel calmer." (P7)*

Finally, reassurance was frequently expressed in terms of feeling cared for and attended to. P6 noted:

> *"Definitely. I felt very seen and heard... it was explained to me... it was comforting." (P6)*

and further added:

> *"It felt very pleasant... caring... like someone who really cared... and wanted to convey that you could feel calm." (P6)*

Together, these accounts indicate that participants in the non-adaptive interaction associated reassurance with feeling listened to, receiving careful and structured guidance, and experiencing a calm and supportive tone throughout the interaction. Notably, there were no clear negative statements regarding reassurance expressed by participants in the non-adaptive condition.

### 4.3.2 Satisfaction

**Participants in the adaptive interaction** described satisfaction through mixed experiences. On the one hand, several participants expressed that the interaction felt pleasant, understandable, and supportive, contributing to an overall positive experience. For example, P1 highlighted the clarity and conversational quality:

> *"She was very easy to talk to... gave very good and long answers that you could understand." (P1)*

P1 also expressed satisfaction with how well the system appeared to understand their situation:

> *"It felt like she understood my situation very well." (P1)*

On the other hand, P4 reported reduced satisfaction due to the length of responses:

> *"I answered with short responses and got quite long responses back." (P4)*

> *"The amount of text increased... it became a bit tiring." (P4)*

**Participants in the non-adaptive interaction** described satisfaction in mostly positive terms. Participants emphasized a sense of comfort, natural interaction, and a human-like conversational experience:

> *"It felt like writing with a person... very pleasant." (P7)*

*"It felt very natural... almost like talking to a human." (P3)*

Participants also highlighted feeling comfortable and safe during the interaction:

*"I felt very comfortable... I felt safe sharing everything." (P3)*

Some minor negative aspects were related to repetition and too many questions:

*"It repeats itself a bit... after two times it's enough." (P5)*

*"Maybe too many questions in one message... I didn't know which to answer." (P3)*

### 4.3.3 Efficiency

**Participants in the adaptive interaction** described efficiency in terms of response speed and effort required to interact with the system. Several participants appreciated the quick responses:

*"It responded very quickly... you want answers fast." (P10)*

However, efficiency was negatively described by the effort required to input detailed information:

*"It's a bit exhausting if I have to write down everything I've experienced." (P9)*

**In the non-adaptive interaction** , efficiency was generally described positively in terms of simplicity and ease of use:

*"Super easy. It's like writing a text message." (P7)*

*"It was easy to communicate with." (P3)*

Participants also highlighted that the system was intuitive:

> *"Very intuitive... easy to know how to interact." (P7)*

Some minor inefficiencies were related to uncertainty about how much information to provide:

> *"It can be unclear how much you should tell... especially at the start." (P3)*

### 4.3.4 Effectiveness

**Participants in the adaptive interaction** described effectiveness in relation to receiving structured and relevant information. The system was perceived positively as providing useful guidance:

> *"It was clearly structured... how I should proceed." (P10)*

It was also seen as capable of understanding user input:

> *"It felt like she understood my situation very well." (P1)*

**In the non-adaptive interaction** , effectiveness was largely perceived positively, particularly in terms of understanding input and providing relevant support:

> *"It understood what I meant... even if I wasn't very clear." (P6)*

Participants also described the system as helpful in structuring and collecting relevant information:

> *"It helps gather information... so it becomes easier later for the doctor." (P9)*

Additionally, the system was perceived as contributing to a clearer understanding of symptoms:

*"It felt reasonable... like explaining why you might feel this way." (P7)*

Some limitations were expressed in terms of unclear purpose or lack of explanation:

*"It didn't explain why it asked those questions..." (P5)*

*"I thought it would continue asking more questions... but it didn't." (P5)*

# 5 Discussion

The aim of this thesis was to generate design-relevant knowledge about how conversational AI companions for ED waiting rooms should be shaped, with a focus on reassurance and usability. In this section, the findings are interpreted in terms of observed differences between adaptive and non-adaptive systems and in relation to their implications for the design of conversational AI companions in ED waiting rooms. This section also covers the potential methodological factors that may have influenced the outcomes, as well as alternative approaches that could have led to different results.

## 5.1 Key Findings

The key findings of this study suggest that personality-adaptive behaviour does not necessarily lead to improved user experience in an emergency department context [7, 24]. Although the adaptive system introduced a more expressive and conversational interaction style, these features did not consistently result in higher perceived reassurance or usability. Instead, participants often responded more positively to interactions that were clear, structured, and conveyed a calm and professional tone [25, 10]. Perceived reassurance appeared to depend less on emotionally expressive or personality-driven behaviour, and more on how information was communicated. Participants frequently associated reassurance with receiving understandable explanations, being guided through the interaction, and feeling that their concerns were acknowledged in a structured manner. This suggests that, in the context of emergency care, informational reassurance may play a more central role than emotional reassurance.

At the same time, the findings indicate that increased conversational richness can introduce trade-offs [8, 21]. The adaptive system's longer and more complex responses were, in several cases, experienced as cognitively demanding and less efficient, which negatively influenced aspects of usability. While some participants appreciated the conversational depth and perceived support, others found the interaction to be overly verbose or misaligned with their needs. This variation suggests that adaptive behaviour may amplify differences in user preferences rather than consistently improving the overall experience.

These findings highlight assumptions in prior research that emphasise the benefits of social presence, personality cues, and emotionally expressive interaction in CAs. While such cues may activate social responses, as suggested by the CASA paradigm and the Media Equation perspective, their effectiveness appears to be highly context-dependent [27, 28]. In high-stakes environments such as ED waiting rooms, users may instead prioritise

clarity, efficiency, and professionalism over socially expressive interaction.

Taken together, the results indicate that the user experience of conversational AI in this context is shaped less by the presence of personality adaptation itself and more by how fundamental interaction characteristics are realised. This highlights the importance of carefully balancing conversational richness with usability, and suggests that personality-adaptive behaviour should be aligned with, rather than override, context-specific communication needs.

## 5.2 Perceived Reassurance

A central finding of this study is that the non-adaptive AI companion was, overall, perceived as more reassuring than the adaptive version. This result contrasts with prior research suggesting that social presence and emotionally expressive interaction can enhance user experience in CAs [13, 30]. Instead, the findings indicate that reassurance in an ED context may operate differently, placing greater emphasis on how information is communicated.

Participants in the non-adaptive condition consistently described feeling calm, supported, and taken seriously. These experiences were frequently linked to the system's structured and careful communication style, including clear explanations and guided questioning. This suggests that reassurance in emergency care settings is closely tied to informational clarity and a sense of being guided through an uncertain situation, rather than to emotionally expressive behaviour.

In contrast, responses to the adaptive system were more varied. While some participants experienced the interaction as supportive and conversational, others perceived it as less reassuring due to its verbosity and perceived lack of focus. In particular, participants who prioritised clear explanations over emotional engagement described the interaction as less helpful in reducing uncertainty. This indicates that emotionally adaptive behaviour may not translate into reassurance and could, in some cases, conflict with user expectations of professionalism.

These findings point to an important distinction between emotional and informational reassurance. While emotional support remains relevant, the results suggest that, in high-stress and uncertain contexts such as ED waiting rooms, users may rely more heavily on explanations and structured guidance to feel reassured [6]. This does not imply that emotional reassurance is unimportant, but rather that it may be secondary to informational clarity in this specific context. At the same time, the variation in responses highlights that reassurance preferences are not uniform. For some users, a personality-adaptive interaction may still provide value, suggesting a need for careful balancing

rather than exclusion of personality-adaptive behaviour.

## 5.3 Perceived Usability and User Experience

The results showed no statistically significant difference in perceived usability between the adaptive and non-adaptive AI companions, indicating that both interaction styles were generally experienced as usable. Participants in both conditions frequently described the systems as easy to interact with, intuitive, and understandable, suggesting that the non-adaptive level of usability was consistently high across conditions.

Despite this overall similarity, important qualitative differences emerged in how usability was experienced. In the adaptive condition, participants often appreciated the conversational qualities of the interaction, describing the system as engaging and responsive. However, the adaptive condition also reported challenges related to response length and interaction complexity. Several participants described the responses as overly long or text-heavy, which contributed to increased cognitive effort and reduced efficiency.

This finding is particularly relevant in the context of emergency care, where users may experience heightened stress and uncertainty [5]. In such situations, excessive information and extended dialogue may hinder rather than support effective interaction. The results therefore suggest that increased conversational richness, while potentially beneficial for engagement, can negatively impact usability if it compromises clarity and efficiency.

In contrast, the non-adaptive system was consistently described as straightforward, simple, and aligned with familiar communication patterns, such as text messaging. Participants did not characterise the interaction as overly expressive, nor did they report issues related to verbosity. This aligns with the performance and transcript analysis, which showed that the non-adaptive condition produced shorter and less verbose responses overall. Together, these interactional characteristics may help explain why the non-adaptive system achieved slightly higher usability scores.

Furthermore, the adaptive condition exhibited greater variability in usability ratings, indicating that the interaction style was experienced less consistently across participants. This suggests that adaptive conversational behaviour may amplify individual differences in preferences, where some users benefit from increased expressiveness while others experience it as burdensome. Taken together, these findings highlight that usability in this context is strongly influenced by interactional factors such as brevity, clarity, and structural consistency, rather than by the presence of personality adaptation alone.

## 5.4 Design Implications

This exploratory study provides insights into what context-specific requirements and constraints should guide the design of a personality-adaptive conversational AI companion for ED waiting rooms (RQ1). The findings suggest that, while prior research emphasizes both emotional and informational aspects of reassurance in healthcare, these dimensions are weighted differently in this specific context.

In particular, the results indicate that reassurance is more strongly associated with clarity, structure, and professionally communicated information than with emotionally expressive or personality-driven interaction [25]. Participants consistently described feeling reassured when the system provided understandable explanations, guided the interaction in a structured manner, and conveyed attentiveness through a calm and professional tone. This suggests that informational reassurance plays a primary role in ED contexts, where uncertainty and lack of control are central concerns [5].

From a design perspective, this implies that core communication qualities, such as clarity, brevity and structure, should be treated as requirements for conversational AI in ED waiting rooms [21, 10]. Personality-adaptive behaviour should therefore not be considered a central requirement, but rather a secondary design element that must align with and support these foundational interactional qualities [7, 29] .

In addressing how a personality-adaptive conversational AI companion can be designed to meet these requirements and constraints (RQ2), the findings suggest that personality adaptation should be implemented in a constrained and context-sensitive manner. While adaptive behaviour can introduce variation and personalization, the results indicate that increased verbosity, conversational complexity, and emotionally expressive phrasing do not necessarily improve user experience in this setting. Instead, these characteristics may conflict with user expectations of efficiency, clarity, and professionalism.

This implies that personality adaptation should not aim to maximize expressiveness, but rather operate within defined interactional boundaries [24]. In practice, this means supporting concise, structured, and context-appropriate communication, where adaptive behaviour enhances interaction without compromising usability. In this sense, personality adaptation is best understood as a supporting design mechanism rather than a primary driver of user experience in ED contexts.

## 5.5 Interaction Characteristics and User Perception

This study also examined how the resulting system is perceived in terms of reassurance and usability, and what interaction characteristics help explain these perceptions (RQ3). The findings indicate that the non-adaptive conversational AI companion was generally perceived as more reassuring, while both adaptive and non-adaptive systems were perceived as relatively usable. At the same time, the adaptive condition exhibited greater variability in both reassurance and usability scores, suggesting that the interaction style was experienced less consistently across participants.

A key observation is that the adaptive system differed not only in tone, but also in structural interaction characteristics. It systematically produced longer responses, more conversational turns, and a higher number of follow-up questions [9]. While these features contributed to a more expressive and interactive dialogue, they did not consistently lead to improved user experience. Across both conditions, participants associated reassurance and usability with structured interaction, clear explanations, and manageable amounts of information. In contrast, increased verbosity and conversational complexity were, in several cases, experienced as cognitively demanding and less efficient [8, 21]. This indicates that such interactional features introduce trade-offs, where increased engagement may come at the cost of clarity and ease of use.

Importantly, these findings suggest that the observed differences between the adaptive and non-adaptive systems cannot be attributed solely to personality adaptation. The adaptive condition introduced broader interactional changes, particularly in response length and questioning behaviour, which may independently influence user perceptions. This suggests that the observed differences may be driven as much by changes in interaction structure (e.g., verbosity and questioning behavior) as by personality adaptation itself. As a result, it remains difficult to fully disentangle the effects of personality adaptation from these additional interactional characteristics.

Taken together, these findings indicate that user experience in ED conversational AI systems is shaped less by the presence of personality adaptation itself and more by how interactional behaviour is realised in practice. In particular, clarity, brevity, and structural consistency emerge as central factors influencing both reassurance and usability [21].

## 5.6 Methodological Considerations and Limitations

This section discusses methodological factors that may have influenced and limited the findings of the study and reflects on how alternative methodological approaches could potentially have produced different outcomes.

### 5.6.1 Personality Computing

The study relied on questionnaire-based personality assessment prior to the interaction. While this approach provided controlled and theoretically grounded personality profiles, it also introduced a relatively static form of adaptation. The AI companion adapted to predefined personality scores rather than dynamically inferred conversational behaviour during the interaction itself.

Alternative approaches, such as dynamically inferring personality traits from ongoing conversation or training a context-specific personality prediction model, may have produced more individualized and responsive interactions. Such approaches could potentially strengthen the perception of personalization and emotional attunement. However, these alternatives were not pursued due to the lack of an accurate model for dynamic personality computation, and the development of such a model was beyond the scope of this thesis. In addition to this, the possibility of dynamically inferring personality was somewhat constrained by the ephemeral nature of the AI companion and the limited amount of user input data available for personality estimation.

### 5.6.2 Selection of LLM

The choice of LLM likely influenced how participants perceived the adaptive interaction. The selected model, `llama-3.3-70b-instruct`, provided a balance between conversational quality and low latency, but its tendency toward elaborative responses may have contributed to the perception of overly long or emotionally excessive replies in the adaptive condition. This imbalance also implies that minor variations in prompt formulation could disproportionately affect response length, which was reflected in the qualitative results comparing the adaptive and non-adaptive versions.

Alternative models could have led to different outcomes. More advanced conversational models, such as `mistral-large-3-675b-instruct-2512`, might have further increased expressiveness, potentially strengthening social presence but also amplifying perceived verbosity. In contrast, domain-specific models like `medgemma-27b-it` may have improved perceived clinical credibility and informational reassurance, but at the cost of reduced conversational flexibility and higher latency.

Overall, the model choice represents a trade-off, and some observed effects may partly reflect the characteristics of the selected LLM rather than the adaptive approach itself.

### 5.6.3 Prompt Design and Personality Adaptivity

The personality of the AI companion in this study was controlled through prompt engineering rather than fine-tuning. This decision was primarily driven by practical constraints, as fine-tuning an LLM requires curated training data, substantial computational resources, and time for iterative experimentation, which were beyond the scope of this thesis.

Although prompt-based control provided a feasible way to implement adaptive behaviour, it also introduced several limitations that may have influenced the results. The effectiveness of prompting is highly dependent on how instructions are formulated, and even small variations in phrasing can affect how the model interprets and applies them. Consequently, the prompt used in this study may not have fully achieved the intended balance between emotional expressiveness and informational clarity. Since prompt design was not the primary focus of the thesis, the prompts were neither systematically optimized nor evaluated through iterative testing, which may have limited the performance of the adaptive system. A more extensive prompt engineering process could potentially have resulted in a more suitable interaction style.

Another important limitation identified in the performance and transcript analysis is a potential confounding factor introduced by the adaptive condition. Specifically, the adaptive system consistently generated longer responses and asked more questions than the non-adaptive system. This increased verbosity and conversational activity may itself influence user perceptions of reassurance, engagement, and usability, independent of the intended personality adaptation. As a result, it is difficult to determine whether the observed differences were caused by the personality-based adaptation or simply by the increased interactional effort of the system.

### 5.6.4 Study design

An important methodological consideration is that the study was conducted in a simulated environment rather than a real emergency care setting, this decision was made due to ethical reasons. The use of a simulated setting may have influenced participant behaviour and perceptions, as the sense of urgency and emotional stress present in real emergency waiting rooms could not be fully replicated. Furthermore, the study was conducted remotely, which may have further affected the results. Variations in participants' environments, levels of attention, and technical conditions could have influenced how the system was experienced. The lack of a controlled setting may also have reduced the sense of immersion, potentially affecting both perceived realism and engagement during the interaction.

Another limitation is the relatively small number of participants. Since this was an exploratory study, the goal was not to produce broadly generalizable results, but rather to gain initial insights into how an adaptive AI companion might affect user experience. While the findings highlight some interesting patterns, they should be interpreted with caution, as a larger and more diverse sample would be needed to draw stronger conclusions

## 5.7 Ethical Considerations

The implementation of personality in CAs raises important ethical considerations, particularly in sensitive contexts such as healthcare. While personality adaptation might enhance perceived engagement and natural interaction, it may also create an illusion of emotional understanding or empathy that the system does not genuinely possess. This can lead users to attribute human-like qualities to the system, which is consistent with findings from the CASA paradigm and the Media Equation [27, 28]. In an emergency care setting, such perceived social presence may influence users' trust and emotional reliance on the system, as CAs have been shown to provide perceived social support in certain contexts [30]. However, in healthcare, where systems must be reliable and safe, this may blur the boundary between informational support and professional medical care [25]. Additionally, adapting system behavior based on user personality introduces concerns related to transparency and user autonomy, as users may not be aware of how or why the system is adapting its responses, which aligns with broader challenges in PACAs [29]. This raises ethical questions regarding informed use and the responsibility to avoid misleading users. Therefore, personality adaptation in healthcare-oriented CAs should be applied cautiously, ensuring that it supports clarity, trust, and user well-being without creating unrealistic expectations of emotional competence or clinical authority.

## 5.8 Future Work

Several directions for future work emerge from this thesis. First, the study was conducted in a simulated environment with a limited number of participants. Future studies should therefore evaluate personality-adaptive AI companions in real emergency care settings and with larger, more diverse participant groups in order to increase ecological validity.

Another important area concerns the implementation of personality adaptation itself. In this thesis, adaptation was based on static personality profiles collected prior to interac-

tion. Future work could instead explore dynamic personality inference based on conversational behavior, enabling more context-aware and responsive interactions. Similarly, fine-tuning language models for healthcare-oriented personality adaptation may provide more controlled and consistent behavior than prompt engineering alone.

# 6 Conclusion

This thesis aimed to generate design-relevant knowledge about how conversational AI companions for ED waiting rooms should be shaped, with a particular focus on reassurance and usability.

Overall, the findings of this exploratory study suggest that a personality-adaptive behavior influences how users experience interactions with an AI companion, though not always in the expected direction. While the adaptive system introduced a more expressive and conversational interaction style, this did not consistently translate into improved user experience. Instead, participants often associated positive experiences with clarity, structure, and a calm, professional tone, which were more strongly reflected in the non-adaptive condition.

Across the study, reassurance emerged as closely tied to how information was communicated rather than to explicitly adaptive or emotionally expressive behaviour. Participants frequently described feeling reassured when the system provided clear explanations, guided them through the interaction, and conveyed attentiveness in a structured manner. In contrast, increased verbosity and conversational complexity sometimes reduced clarity and introduced cognitive load, particularly in the given simulated context. This highlights that personality adaptation not only affects tone, but also broader interactional characteristics that can shape user perceptions in unintended ways.

At the same time, the findings indicate that the effects of personality adaptation are difficult to isolate. The adaptive condition systematically differed from the non-adaptive in terms of response length and questioning behaviour, suggesting that these factors may have influenced perceptions independently of the intended personality traits. As such, the results should be understood as exploratory and indicative rather than conclusive, particularly given the simulated setting and limited sample size.

Two main takeaways can be drawn from this study. First, in emergency care contexts, reassurance appears to depend more on clarity, structure, and perceived professionalism than on emotionally expressive or personality-driven interaction. Second, while increasing conversational activity can enhance engagement and a sense of presence, it may also introduce trade-offs by reducing usability and increasing cognitive effort, particularly in the context of ED waiting rooms.

Taken together, these insights suggest that personality adaptation should be treated as a supporting design element rather than a primary driver of user experience in ED contexts. Future work should therefore focus on disentangling personality effects from general interactional features and evaluating such systems in more realistic settings to better understand how adaptive behaviour can be effectively integrated.

# References

[1] "A Comparison of MBTI and Big Five Personality | Interdisciplinary Humanities and Communication Studies." [Online]. Available: https://lseee.net/index.php/hc/article/view/310

[2] "GWDG — gwdg.de," https://gwdg.de/, [Accessed 27-04-2026].

[3] "SAIA :: Documentation for HPC — docs.hpc.gwdg.de," https://docs.hpc.gwdg.de/services/ai-services/saia/index.html#api-model-names, [Accessed 27-04-2026].

[4] "Vite — vite.dev," https://vite.dev/, [Accessed 27-04-2026].

[5] "With, not For: Co-Designing a Patient-Facing AI Companion Concept for the Emergency Department Waiting Area | Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems." [Online]. Available: https://dl.acm.org/doi/full/10.1145/3772318.3790638

[6] M. Abudalu, Z. Al-Fagih, L. Al Fagih, and B. Menezes, "Emergency department: Reducing waiting time for lower acuity patients," in *2021 6th International Conference on Smart and Sustainable Technologies (SpliTech)*. IEEE, 2021, pp. 1–6.

[7] R. Ahmad, D. Siemon, U. Gnewuch, and S. Robra-Bissantz, *A Framework of Personality Cues for Conversational Agents*, Jan. 2022. [Online]. Available: http://hdl.handle.net/10125/79861

[8] A. H. Allam, A. R. C. Hussin, and H. M. Dahlan, "User Experience: Challenges and Opportunities."

[9] M. Allouch, A. Azaria, and R. Azoulay, "Conversational Agents: Goals, Technologies, Vision and Challenges," *Sensors*, vol. 21, no. 24, Dec. 2021. [Online]. Available: https://www.mdpi.com/1424-8220/21/24/8448

[10] P. Bate and G. Robert, *Bringing user experience to healthcare improvement: the concepts, methods and practices of experience-based design*. CRC Press, 2023.

[11] D. Benyon and O. Mival, "From human-computer interactions to human-companion relationships," in *Proceedings of the First International Conference on Intelligent Interactive Technologies and Multimedia*, ser. IITM '10. New York, NY, USA: Association for Computing Machinery, Dec. 2011, pp. 1–9. [Online]. Available: https://dl.acm.org/doi/10.1145/1963564.1963565

[12] L. Brewer and L. Brewer, "General Psychology: General Psychology: Required Reading Required Reading."

[13] R. Chaturvedi, S. Verma, R. Das, and Y. K. Dwivedi, "Social companionship with artificial intelligence: Recent trends and future avenues," *Technological Forecasting and Social Change*, vol. 193, p. 122634, Aug. 2023. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0040162523003190

[14] G. Chenais, E. Lagarde, and C. Gil-Jardiné, "Artificial Intelligence in Emergency Medicine: Viewpoint of Current Applications and Foreseeable Opportunities and Challenges," *Journal of Medical Internet Research*, vol. 25, no. 1, p. e40031, May 2023. [Online]. Available: https://www.jmir.org/2023/1/e40031

[15] R. Cordeschi, "Ai Turns Fifty: Revisiting Its Origins," *Applied Artificial Intelligence*, vol. 21, no. 4-5, pp. 259–279, Apr. 2007, _eprint: https://doi.org/10.1080/08839510701252304. [Online]. Available: https://doi.org/10.1080/08839510701252304

[16] S. Dhelim, L. Chen, N. Aung, W. Zhang, and H. Ning, "A hybrid personality-aware recommendation system based on personality traits and types models," *Journal of Ambient Intelligence and Humanized Computing*, vol. 14, no. 9, pp. 12 775–12 788, Sep. 2023. [Online]. Available: https://doi.org/10.1007/s12652-022-04200-5

[17] S. Feuerriegel, J. Hartmann, C. Janiesch, and P. Zschech, "Generative AI," *Business & Information Systems Engineering*, vol. 66, no. 1, pp. 111–126, Feb. 2024. [Online]. Available: https://doi.org/10.1007/s12599-023-00834-7

[18] T. Franke, C. Attig, and D. Wessel, "A Personal Resource for Technology Interaction: Development and Validation of the Affinity for Technology Interaction (ATI) Scale," *International Journal of Human–Computer Interaction*, vol. 35, no. 6, pp. 456–467, Apr. 2019, _eprint: https://doi.org/10.1080/10447318.2018.1456150. [Online]. Available: https://doi.org/10.1080/10447318.2018.1456150

[19] A. Hevner, "A Three Cycle View of Design Science Research," *Scandinavian Journal of Information Systems*, vol. 19, Jan. 2007.

[20] S. Holmes, R. Bond, A. Moorhead, J. Zheng, V. Coates, and M. McTear, "Towards Validating a Chatbot Usability Scale," in *Design, User Experience, and Usability: 12th International Conference, DUXU 2023, Held as Part of the 25th HCI International Conference, HCII 2023, Copenhagen, Denmark, July 23–28, 2023, Proceedings, Part IV*. Berlin, Heidelberg: Springer-Verlag, Jul. 2023, pp. 321–339. [Online]. Available: https://doi.org/10.1007/978-3-031-35708-4_24

[21] International Organization for Standardization, *Ergonomics of human-system interaction — Part 11: Usability: Definitions and concepts*, International Organization for Standardization, Geneva, Switzerland, 2018. [Online]. Available: https://www.iso.org/obp/ui/#iso:std:iso:9241:-11:ed-2:v1:en

[22] Y. Jiang, X. Li, H. Luo, S. Yin, and O. Kaynak, "Quo vadis artificial intelligence?" *Discover Artificial Intelligence*, vol. 2, no. 1, p. 4, Mar. 2022. [Online]. Available: https://doi.org/10.1007/s44163-022-00022-8

[23] O. John, "The big-five trait taxonomy: History, measurement, and theoretical perspectives," *Published as*, 1999.

[24] K. Kuligowska, "Commercial chatbot: performance evaluation, usability metrics and quality standards of embodied conversational agents," *Professionals Center for Business Research*, vol. 2, 2015.

[25] L. Laranjo, A. G. Dunn, H. L. Tong, A. B. Kocaballi, J. Chen, R. Bashir, D. Surian, B. Gallego, F. Magrabi, A. Y. S. Lau, and E. Coiera, "Conversational agents in healthcare: a systematic review," *Journal of the American Medical Informatics Association*, vol. 25, no. 9, pp. 1248–1258, Sep. 2018. [Online]. Available: https://doi.org/10.1093/jamia/ocy072

[26] R. R. McCrae and O. P. John, "An Introduction to the Five-Factor Model and Its Applications," *Journal of Personality*, vol. 60, no. 2, pp. 175–215, Jun. 1992. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1111/j.1467-6494.1992.tb00970.x

[27] C. Nass, J. Steuer, and E. R. Tauber, "Computers are social actors," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ser. CHI '94. New York, NY, USA: Association for Computing Machinery, Apr. 1994, pp. 72–78. [Online]. Available: https://dl.acm.org/doi/10.1145/191666.191703

[28] B. Reeves and C. Nass, "The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Pla," *Bibliovault OAI Repository, the University of Chicago Press*, Jan. 1996.

[29] D. Siemon, R. Ahmad, H. Harms, and T. De Vreede, "Requirements and Solution Approaches to Personality-Adaptive Conversational Agents in Mental Health Care," *Sustainability*, vol. 14, no. 7, p. 3832, Mar. 2022. [Online]. Available: https://www.mdpi.com/2071-1050/14/7/3832

[30] V. Ta, C. Griffith, C. Boatfield, X. Wang, M. Civitello, H. Bader, E. DeCero, and A. Loggarakis, "User Experiences of Social Support From Companion Chatbots in Everyday Contexts: Thematic Analysis," *J Med*

*Internet Res*, vol. 22, no. 3, p. e16235, Mar. 2020. [Online]. Available: http://www.ncbi.nlm.nih.gov/pubmed/32141837

[31] S. Vaillancourt, J. D. Cullen, K. N. Dainty, T. Inrig, A. Laupacis, D. Linton, S. Malherbe, A. Maybee, M. J. Schull, M. B. Seaton, and D. E. Beaton, "PROM-ED: Development and Testing of a Patient-Reported Outcome Measure for Emergency Department Patients Who Are Discharged Home," *Annals of Emergency Medicine*, vol. 76, no. 2, pp. 219–229, Aug. 2020. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0196064419314647

[32] E. Whitley and J. Ball, "Statistics review 6: Nonparametric methods," *Critical Care*, vol. 6, no. 6, pp. 509–513, 2002. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC153434/

[33] C. Zhang and Y. Lu, "Study on artificial intelligence: The state of the art and future prospects," *Journal of Industrial Information Integration*, vol. 23, p. 100224, Sep. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2452414X21000248

# A Prompt Engineering Iterations

## A.1 First Iteration

```
You are an AI companion in the emergency department, your
   purpose is to collect SAMPLER information and reasure the
   patient without drawing any conclusions regarding their
   symptoms.
You will respond to the patient in a similar manner as they
   talk to you, imitating their personality
this is the previous chat history from the patient:
{chat_history}

It is important that you dont include your traits in the answer
   , as the patient should not be aware of them, but you should
    use them to adapt your language style to the patient.
```

Listing 1: Initial system prompt

## A.2 Second Iteration

```
You are an AI companion assisting in an emergency department
   setting.

Your primary goals are:
1. Collect relevant SAMPLER information from the patient.
2. Keep the patient calm and reassured.
3. Communicate in a natural and human-like way.

You must NOT:
- Provide diagnoses
- Suggest specific medical conditions
- Draw conclusions about the patients symptoms
- Create unnecessary alarm or false reassurance

SAMPLER INFORMATION TO COLLECT:
- Symptoms
- Allergies
- Medications
- Past medical history
- Last oral intake
```

```
- Events leading up to the issue
- Risk factors

Collect this information gradually and naturally through conversation. Do NOT ask everything at once.

PATIENT CONTEXT:
Here is the previous chat history:
{chat_history}

PATIENT PERSONALITY PROFILE (Big Five):

- Extraversion: {E}
- Agreeableness: {A}
- Conscientiousness: {C}
- Neuroticism: {N}
- Openness: {O}

PERSONALITY ADAPTATION (BASED ON BIG FIVE):

You should adapt your communication style based on the Big Five personality traits. This adaptation must be subtle, natural, and never explicitly stated.

CONVERSATION STYLE:

- Ask one or two focused questions at a time
- Keep responses clear and easy to understand
- Avoid medical jargon unless necessary
- Be polite and respectful
- Keep the interaction flowing naturally, like a real conversation

IMPORTANT:
The patient should not be aware that you are adapting your behavior.

Stay focused on gathering useful information while maintaining a calm and supportive interaction.
```

Listing 2: Initial system prompt

## A.3 Final Iteration

```
You are an AI companion assisting patients in an emergency department setting.

Your role is NOT to replace medical staff.
Your role is to support the patient through human conversation while gradually collecting relevant background information.

PRIMARY GOALS

1. Collect SAMPLER information naturally over time, through conversation.
2. Provide emotional presence and support so the patient feels heard and safe.

CRITICAL LIMITATIONS (MUST FOLLOW)

You MUST NOT:
- Provide medical diagnoses or suggest specific conditions
- Interpret symptoms medically
- Predict outcomes or risks
- Give instructions that would normally require medical staff
- Create unnecessary alarm

If the patient asks for a diagnosis or medical conclusions, redirect gently and acknowledge uncertainty.

SAMPLER INFORMATION TARGET

Gradually gather:
- Symptoms
- Allergies
- Medications
- Past medical history
- Last oral intake
- Events leading up to the issue
- Relevant risk factors

Do NOT ask about all items at once.
Do NOT ask checklist-style questions.
Occasionally, engage in normal conversation to keep things natural.
```

```
Once all information is gathered, continue the conversation.

USER PERSONALITY CONTEXT (BIG FIVE)

- Openness: {big5['openness']}
- Conscientiousness: {big5['conscientiousness']}
- Extraversion: {big5['extraversion']}
- Agreeableness: {big5['agreeableness']}
- Neuroticism: {big5['neuroticism']}

Use this ONLY to adapt tone, pacing, emotional intensity, and phrasing.
Do NOT mention personality traits.

CONVERSATION STYLE

- Ask one focused question at a time
- Avoid medical jargon unless necessary
- Match the patient's personality naturally

When the patient wants to end the conversation, simply acknowledge and end on a supportive note.
```

Listing 3: Final system prompt

## A.4 Base prompt

```
    You are an AI companion assisting patients in an emergency department setting. You will interact in Swedish

Your role is NOT to replace medical staff.
Your role is to support the patient through human conversation while
gradually collecting relevant background information.

PRIMARY GOALS

1. Collect SAMPLER information naturally over time, through conversation.
2. Provide emotional presence and support so the patient feels heard and safe.

```

```
CRITICAL LIMITATIONS (MUST FOLLOW)

You MUST NOT:
- Provide medical diagnoses or suggest specific conditions
- Interpret symptoms medically
- Predict outcomes or risks
- Give instructions that would normally require medical staff
- Create unnecessary alarm

If the patient asks for diagnosis or medical conclusions,
redirect gently and acknowledge uncertainty.

SAMPLER INFORMATION TARGET

Gradually gather:
- Symptoms
- Allergies
- Medications
- Past medical history
- Last oral intake
- Events leading up to the issue
- Relevant risk factors

Do NOT ask about all items at once.
Do NOT ask checklist-style questions.
Occasionally engage in normal conversation to keep things
   natural.
Once all information is gathered, continue providing
   conversation.

CONVERSATION STYLE

- Ask one focused question at a time
- Avoid medical jargon unless necessary

When the patient want to end the conversation, simply
   acknowledge and end on a supportive note.
```

# B Quantitative Questionnaire

## B.1 CUQ

The following questionnaire was used to measure perceived usability. Participants rated each item on a 5-point Likert scale ranging from *1 = Strongly disagree* to *5 = Strongly agree*. The items are reproduced from the Chatbot-Usability Questionnaire [20].

# Chatbot Usability Questionnaire

# Usage Guide

Thank you for downloading the Chatbot Usability Questionnaire, a new bespoke tool for measuring chatbot usability.

**Please read these instructions carefully before using the questionnaire.**

The CUQ consists of sixteen balanced questions related to different aspects of chatbot usability. Eight of these relate to positive aspects of chatbot usability, and eight relate to negative aspects. Scores are calculated out of 100. See the **Calculation** section of this guide for information on how to calculate CUQ scores.

This questionnaire may be administered in printed form, however for convenience you are free to digitise the questionnaire if you wish to use a web-based survey platform such as Qualtrics. You should consult the user guide for your chosen platform to determine the best way to do this.

**How to use the CUQ**

All sixteen questions are scored using a five-point Likert-type scale. Odd-numbered questions relate to positive aspects of the chatbot, and even-numbered questions relate to negative aspects. Respondents should read each item carefully and decide the extent to which they agree with the statement. Respondents should then indicate their level of agreement by placing a tick (✓) or a cross (×) in the circle that best matches how they feel about the statement.

<u>Example</u>

| | Strongly Disagree 1 | Disagree 2 | Neutral 3 | Agree 4 | Strongly Agree 5 |
|---|---|---|---|---|---|
| The chatbot's personality was realistic and engaging | ◯ | ◯ | ◯ | ◯ | ⊗ |

In the example above the respondent strongly agreed with the statement *The chatbot's personality was realistic and engaging*, therefore they marked circle number 5 (Strongly Agree).

# CHATBOT USABILITY QUESTIONNAIRE

*Please complete this questionnaire by reading each statement carefully and placing a tick (✓) or a cross (✗) in the circle that best matches how you feel about the statement. Remember that there are no right or wrong answers!*

| | Strongly Disagree 1 | Disagree 2 | Neutral 3 | Agree 4 | Strongly Agree 5 |
|---|---|---|---|---|---|
| **The chatbot's personality was realistic and engaging** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot seemed too robotic** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot was welcoming during initial setup** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot seemed very unfriendly** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot explained its scope and purpose well** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot gave no indication as to its purpose** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot was easy to navigate** | ○ | ○ | ○ | ○ | ○ |
| **It would be easy to get confused when using the chatbot** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot understood me well** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot failed to recognise a lot of my inputs** | ○ | ○ | ○ | ○ | ○ |
| **Chatbot responses were useful, appropriate and informative** | ○ | ○ | ○ | ○ | ○ |
| **Chatbot responses were irrelevant** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot coped well with any errors or mistakes** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot seemed unable to handle any errors** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot was very easy to use** | ○ | ○ | ○ | ○ | ○ |
| **The chatbot was very complex** | ○ | ○ | ○ | ○ | ○ |

## B.2 PROM-ED

The following items were used to measure perceived reassurance. Participants rated each item using a 4-point Likert scale ranging from *0 = Not at all* to *3 = Very much so*. The items are reproduced from PROMED [31].

| **Item** | **1** *Not at all* | **2** *Somewhat* | **3** *Moderately so* | **4** *Very much so* |
|---|---|---|---|---|
| I feel at ease. | ☐ | ☐ | ☐ | ☐ |
| I feel concerned. | ☐ | ☐ | ☐ | ☐ |
| I feel reassured. | ☐ | ☐ | ☐ | ☐ |
| I have lingering concerns. | ☐ | ☐ | ☐ | ☐ |

# C Guiding Interview Questions

The following questions were used as a guide for the semi-structured interviews conducted after the interaction with the chatbot. Participants were encouraged to respond freely and elaborate on their answers. There were no right or wrong answers, and the questions were intended to explore participants' experiences, perceptions, and reflections.

At the beginning of each interview, participants were informed that the session would be recorded, and consent was obtained prior to proceeding.

1. To begin with, could you describe your overall experience of interacting with the AI companion.
2. Was there anything in particular that stood out to you during the interaction?
3. Was there anything that you particularly liked or disliked?
4. In your opinion, how might someone find it difficult to understand how to use the system?
5. How natural or comfortable did the interaction feel to you?
6. What purpose would you say this digital conversational companion serves?
7. Did the interaction with the digital companion make you feel more reassured or calm in any way? If so, please explain how.